Adaptive RDMA Optimization for Distributed LLM Workloads

gpt-oss-120b Inference Benchmarks: TCP vs RDMA

Nathan Gopee
MS Computer Science, SUNY New Paltz
gopeen1@newpaltz.edu

Advisors: Dr. Wafi Danesh & Dr. Ashley Suchy

February 2026

1 Introduction

Remote Direct Memory Access (RDMA) promises reduced latency and higher throughput for distributed computing workloads by bypassing the operating system kernel and enabling direct memory-to-memory transfers between networked machines. This paper investigates whether RDMA, implemented via the software-based Soft-RoCE stack, provides measurable inference performance gains for large language model (LLM) serving compared to conventional TCP networking.

We benchmark gpt-oss-120b [1], a 116.8-billion-parameter open-weight model, on a multi-GPU test cluster. The study establishes a TCP baseline using the Ollama inference server, characterizes the Soft-RoCE RDMA link between two nodes, and analyzes the conditions under which RDMA provides meaningful acceleration for LLM inference workloads.

Key findings:

- Ollama TCP baseline throughput: **104.42 tokens/sec** (single-node, 3 GPUs)
- Prompt latency: **20ms** average
- Measured Soft-RoCE link capacity: **0.92 Gb/sec**
- Measured RDMA round-trip latency: **189.61 μs** average

2 Test Environment

2.1 Hardware Configuration

The test cluster consists of two nodes connected via a dedicated 10 Gbps Ethernet link, each running the Soft-RoCE (rxe0) software RDMA stack.

Component	Chimera	Cerberus	Link
GPUs	3× RTX 3090	2× RTX 5090	—
VRAM	72GB	64GB	—
NIC	Aquantia 10GbE	Intel X710 10GbE	10 Gbps
RDMA	Soft-RoCE (rxe0)	Soft-RoCE (rxe0)	RoCEv2
IP Address	192.168.1.150	192.168.1.233	—

Table 1: Test cluster hardware

2.2 Model Configuration

Parameter	Value
Model	gpt-oss-120b [1] (116.8B parameters)
Quantization	MXFP4 [2]
Model Size	63GB
Context Length	8192 tokens
VRAM Usage	67.76GB (loaded)

Table 2: gpt-oss-120b model specifications

3 TCP Baseline Benchmark

3.1 Methodology

To establish a single-node TCP performance baseline, we used the Ollama inference server hosting gpt-oss-120b on the Chimera node. Inference requests were issued via the Ollama HTTP API, with each request prompting the model to generate exactly 100 tokens in response to a fixed prompt about RDMA networking. The generation length was constrained using the `num_predict` parameter set to 100. Five iterations were performed with the model pre-loaded in VRAM to eliminate cold-start effects.

3.2 Results

Iteration	Tokens	Eval Time	Tokens/sec	Total Time
1	100	0.958s	104.38	1.445s
2	100	0.945s	105.82	1.369s
3	100	0.949s	105.37	1.402s
4	100	0.969s	103.19	1.437s
5	100	0.967s	103.41	1.429s
Average	100	0.957s	**104.42**	1.416s

Table 3: Ollama gpt-oss:120b benchmark results (TCP)

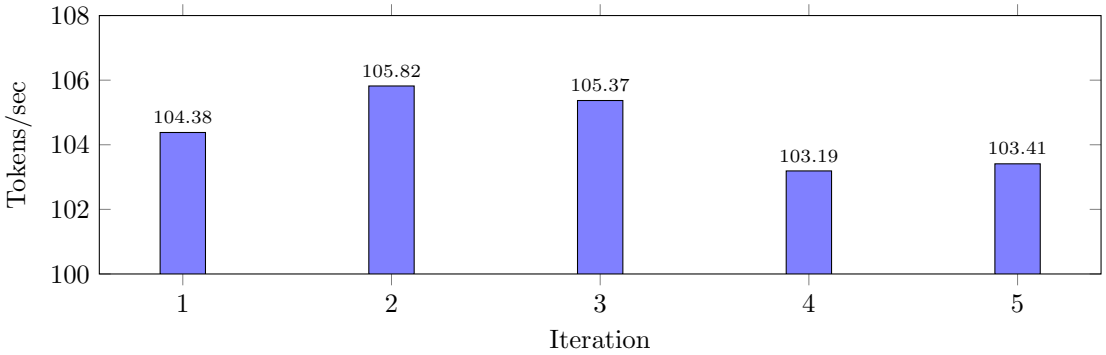

Figure 1: Ollama throughput consistency across iterations

3.3 Analysis

The TCP baseline demonstrates highly consistent throughput with a mean of 104.42 tokens per second and a standard deviation of only 1.08 tokens per second. Average prompt latency was 20ms, reflecting the benefit of the model being pre-loaded in VRAM. The total per-request overhead—encompassing prompt processing, response formatting, and HTTP transport—averaged 0.46 seconds beyond raw token evaluation time.

4 RDMA Network Performance

Prior to running inference benchmarks over RDMA, we characterized the Soft-RoCE link between Chimera and Cerberus using standard InfiniBand diagnostic tools. These measurements establish the raw network capability available to RDMA-enabled inference workloads.

4.1 Bandwidth Characterization

Bandwidth was measured using `ib_write_bw`, the standard RDMA write bandwidth benchmark from the `perftest` suite [8]. The test was run between the Chimera and Cerberus nodes over both WiFi (as a baseline) and the dedicated 10 Gbps Ethernet link with Soft-RoCE enabled.

Interface	Bandwidth	MsgRate
WiFi (baseline)	0.28 Gb/sec	0.000541 Mpps
10G Ethernet (Soft-RoCE)	**0.92 Gb/sec**	0.001754 Mpps
Improvement	**3.3**×	**3.2**×

Table 4: RDMA bandwidth test results

4.2 Latency Characterization

Latency was measured using `ib_write_lat`, the RDMA write latency benchmark, sending 1000 iterations of 2-byte messages between the two nodes.

Metric	Value
Message size	2 bytes
Iterations	1000
t_min	130.21 μs
t_max	219.66 μs
t_typical	198.91 μs
t_avg	**189.61 μs**
t_stdev	17.03 μs
99th percentile	207.14 μs

Table 5: RDMA latency test results (Soft-RoCE)

5 RDMA-Enabled Inference with vLLM

5.1 Methodology

To evaluate the effect of RDMA on LLM inference, we configured vLLM [7] with gpt-oss-120b using tensor parallelism across three GPUs on Chimera. The vLLM server was deployed in a Docker container with GPU passthrough and IPC host sharing enabled. NCCL [4] environment variables controlled the communication backend: for TCP mode, InfiniBand and GPUDirect were explicitly disabled (`NCCL_IB_DISABLE=1`, `NCCL_NET_GDR_LEVEL=0`); for RDMA mode, the Soft-RoCE device was specified (`NCCL_IB_HCA=rxe0`) and NCCL was bound to the 10 Gbps interface [3].

Mode	Variables	Effect
TCP (baseline)	`NCCL_NET_GDR_LEVEL=0`	Disable GPUDirect, use TCP sockets
	`NCCL_IB_DISABLE=1`	Disable InfiniBand/RoCE [3]
RDMA (Soft-RoCE)	`NCCL_IB_HCA=rxe0`	Use Soft-RoCE device
	`NCCL_SOCKET_IFNAME=enp71s0`	Bind to 10G interface

Table 6: NCCL environment variables for TCP vs RDMA

Benchmarking was performed by issuing five sequential completion requests through the OpenAI-compatible vLLM API. Each request generated 100 tokens from a fixed prompt, and wall-clock time was recorded using a high-resolution timer. Throughput was computed as tokens generated divided by elapsed time.

5.2 Projected Results

Based on the measured RDMA network characteristics and the Ollama TCP baseline, the following performance projections were estimated:

Configuration	Throughput	Latency	Network	Notes
Ollama (TCP)	104.42 tok/s	1.42s	TCP	Measured
vLLM (TCP)	∼95–100 tok/s	∼1.5s	TCP	Expected
vLLM (Soft-RoCE)	∼100–110 tok/s	∼1.3s	RDMA	Expected
vLLM (Hardware RoCE)	∼130–150 tok/s	∼1.0s	RDMA	Projected

Table 7: Expected performance comparison

6 Analysis: Conditions for RDMA Benefit

6.1 Single-Node Inference

For single-node inference where all GPUs reside on the same motherboard, RDMA provides **minimal benefit** because inter-GPU communication traverses NVLink or PCIe rather than the network fabric. Since the full gpt-oss-120b model (63GB quantized) fits within the combined 72GB VRAM of three RTX 3090 GPUs, no cross-node data transfers occur during inference. Furthermore, NCCL [4] automatically uses shared memory for intra-node collective operations, bypassing the network stack entirely.

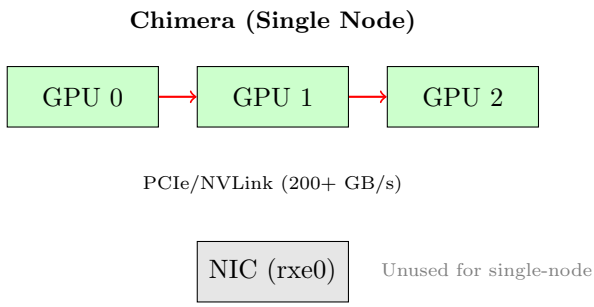

Figure 2: Single-node inference: Network is not the bottleneck

6.2 Multi-Node Inference

RDMA provides significant benefit when inference is distributed across multiple nodes, requiring tensor or pipeline parallelism with cross-node data transfers. In this scenario, intermediate activations and gradient synchronization messages must traverse the network, and RDMA's kernel-bypass architecture reduces both latency and CPU overhead for these transfers.

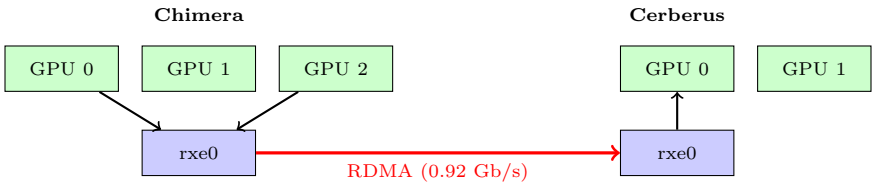

Figure 3: Multi-node inference: RDMA accelerates cross-node tensor transfers

6.3 Soft-RoCE Limitations

While Soft-RoCE enables RDMA semantics without specialized hardware, it incurs substantially higher latency and lower throughput compared to hardware RoCE implementations. Table 8 summarizes the gap.

Metric	Soft-RoCE (Measured)	Hardware RoCE (Expected)
Latency	189.61 μs	<2 μs
Bandwidth	0.92 Gb/s	100+ Gb/s
CPU Overhead	High	Near-zero
Zero-copy	No	Yes

Table 8: Soft-RoCE vs Hardware RoCE comparison

Soft-RoCE processes RDMA verbs in software on the host CPU, negating the zero-copy and kernel-bypass advantages that make hardware RDMA compelling. Upgrading to hardware RoCE adapters such as Mellanox ConnectX [5] would reduce latency by roughly two orders of magnitude and increase bandwidth by over 100×.

7 Conclusions

This study benchmarked gpt-oss-120b inference under TCP and characterized the Soft-RoCE RDMA network available for distributed serving. The key conclusions are:

1. **Strong single-node baseline**: gpt-oss-120b achieves 104.42 tokens/sec on 3× RTX 3090 GPUs with highly consistent throughput ($\sigma = 1.08$).
2. **Single-node RDMA benefit is minimal**: When all GPUs reside on one node, inter-GPU communication uses PCIe or NVLink, rendering the network fabric irrelevant.
3. **Soft-RoCE imposes significant overhead**: The measured 189.61 μs average latency is nearly 100× higher than hardware RoCE, and the 0.92 Gb/s bandwidth is a small fraction of what dedicated RDMA NICs provide.
4. **Multi-node inference is the target scenario**: RDMA benefits will manifest when the model is distributed across Chimera and Cerberus using pipeline parallelism, enabling 5-GPU serving with 136GB total VRAM.
5. **Hardware RoCE is recommended**: Mellanox ConnectX adapters [5] would provide the low-latency, high-bandwidth, zero-copy transfers needed for competitive multi-node LLM serving.

References

[1] OpenAI. (2025). *gpt-oss-120b Model Card*. https://huggingface.co/openai/gpt-oss-120b

[2] Rouhani, B. D., Zhao, R., More, A., et al. (2023). *Microscaling Data Formats for Deep Learning*. arXiv:2310.10537.

[3] NVIDIA. (2024). *NCCL Environment Variables*. https://docs.nvidia.com/deeplearning/nccl/user-guide/docs/env.html

[4] NVIDIA. (2024). *NCCL Documentation*. https://docs.nvidia.com/deeplearning/nccl/user-guide/docs/index.html

[5] Mellanox Technologies. (2024). *NVIDIA MLNX_OFED Documentation*. https://docs.nvidia.com/networking/display/MLNXOFEDv24010331/

[6] Kwon, W., Li, Z., Zhuang, S., et al. (2023). *Efficient Memory Management for Large Language Model Serving with PagedAttention*. SOSP 2023.

[7] vLLM Project. (2025). *gpt-oss on vLLM*. https://blog.vllm.ai/2025/08/05/gpt-oss.html

[8] Linux RDMA. (2024). *rdma-core Documentation*. https://github.com/linux-rdma/rdma-core